

PE-EC702A — Adaptive Signal Processing

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Module 3 Exam Notes: LMS Variants & Signal Space Concepts

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1. Variants of the LMS Algorithm

To overcome constraints of standard LMS (computational complexity, time-varying power instability, or convergence rates), several variants are employed.

1.1 The Sign-LMS Family

Designed to reduce computational requirements in high-speed hardware by replacing multiplications with sign operations. The vector sign function $\text{sgn}(\cdot)$ extracts element-wise signs (± 1 or 0).

1. **Sign-Error LMS:** Simplifies the error term.

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \text{sgn}(e(n))\mathbf{x}(n) \quad (1)$$

2. **Sign-Data (Regressor) LMS:** Simplifies the regressor data vector.

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n) \text{sgn}(\mathbf{x}(n)) \quad (2)$$

3. **Sign-Sign LMS:** Simplifies both terms; updates require only additions (no multiplications).

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \text{sgn}(e(n)) \text{sgn}(\mathbf{x}(n)) \quad (3)$$

1.2 Normalized LMS (NLMS) Algorithm

In standard LMS, the step-size is bounded by the signal power ($0 < \mu < 2/\text{Tr}(\mathbf{R})$). If the input power varies widely (like speech), standard LMS can diverge during high-power segments. NLMS solves this by normalizing the step-size with the instantaneous squared norm of the input vector:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\tilde{\mu}}{\|\mathbf{x}(n)\|^2 + \epsilon} e^*(n) \mathbf{x}(n) \quad (4)$$

where $\tilde{\mu}$ is the normalized step-size ($0 < \tilde{\mu} < 2$), and $\epsilon > 0$ is a small regularization parameter preventing division by zero.

1.3 Block LMS (BLMS) and FFT-Based Realizations

Standard LMS updates weights sample-by-sample. Block LMS updates the weights only once per block of L input samples, averaging the gradients over the block:

$$\mathbf{w}(k+1) = \mathbf{w}(k) + \frac{2\mu}{L} \sum_{i=0}^{L-1} e(kL+i) \mathbf{x}(kL+i) \quad (5)$$

where k is the block index.

- **FFT Realization (FDAF):** Block convolution and correlation can be computed in the frequency domain using Fast Fourier Transforms (FFTs) via the overlap-save or overlap-add method. This reduces computational complexity from $O(M^2)$ to $O(M \log_2 M)$ for large tap lengths.

1.4 Sub-band Adaptive Filtering

In sub-band adaptive filters, the wideband input and desired signals are split into N sub-bands using an analysis filter bank. Adaptive filtering is performed on each sub-band signal after decimation (downsampling). Finally, a synthesis filter bank reconstructs the full-band output.

- **Advantages:** Decimation reduces the operating sampling rate. Sub-band signals have a much whiter spectral envelope (smaller eigenvalue spread) than the original wideband signal, significantly accelerating convergence.

2. Signal Space Concepts

To formulate advanced algorithms like RLS, signals are viewed as elements of abstract vector spaces (Hilbert spaces).

2.1 Vector Space and Subspace Theory

A vector space V over a field F (typically complex numbers $\mathbb{C}$) is a set of elements closed under vector addition and scalar multiplication, satisfying the standard vector space axioms.

- **Subspace:** A subset $W \subseteq V$ is a subspace if it is closed under vector addition and scalar multiplication.
- **Basis and Dimension:** A set of vectors $\{v_1, \dots, v_k\}$ forms a basis for V if they are linearly independent and span V . The number of vectors in the basis is the dimension $\dim(V)$.
- **Linear Operators, Rank and Nullity:** For a linear operator $T : V \rightarrow W$, the range space is $\mathcal{R}(T)$ and null space is $\mathcal{N}(T)$. The dimensions are $\text{rank } \rho(T) = \dim(\mathcal{R}(T))$ and nullity $\nu(T) = \dim(\mathcal{N}(T))$. The Rank-Nullity Theorem states:

$$\rho(T) + \nu(T) = \dim(V) \quad (6)$$

2.2 Inner Product Space and Orthogonality

An inner product space is a vector space V equipped with an inner product $\langle u, v \rangle$, mapping two vectors to a scalar, satisfying conjugate symmetry ($\langle u, v \rangle = \langle v, u \rangle^*$), linearity, and positive definiteness ($\langle u, u \rangle \geq 0$).

- **Induced Norm:** $\|u\| = \sqrt{\langle u, u \rangle}$.
- **Orthogonality:** Two vectors are orthogonal if $\langle u, v \rangle = 0$.
- **Orthogonal Decomposition:** Any vector space V with a closed subspace W can be decomposed uniquely as:

$$V = W \oplus W^\perp \quad (7)$$

where $W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \ \forall w \in W\}$ is the orthogonal complement.

2.3 Gram-Schmidt Orthogonalization

A recursive algorithm that converts a set of linearly independent vectors $\{v_1, v_2, \dots, v_M\}$ into an orthonormal set $\{q_1, q_2, \dots, q_M\}$ spanning the same space:

$$\begin{aligned} u_1 &= v_1, \quad q_1 = \frac{u_1}{\|u_1\|} \\ u_k &= v_k - \sum_{j=1}^{k-1} \langle v_k, q_j \rangle q_j, \quad q_k = \frac{u_k}{\|u_k\|} \quad \text{for } k = 2, \dots, M \end{aligned} \quad (8)$$

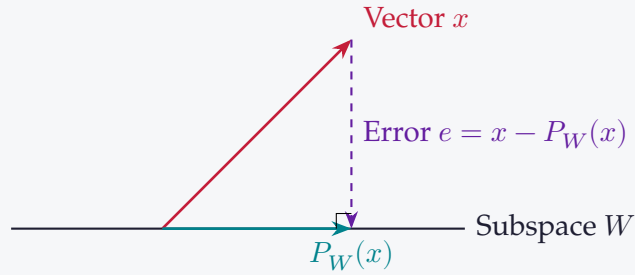
2.4 Orthogonal Projection

The orthogonal projection of a vector x onto a subspace W spanned by orthonormal vectors $\{q_1, \dots, q_k\}$ is:

$$P_W(x) = \sum_{i=1}^k \langle x, q_i \rangle q_i \quad (9)$$

The projection error $e = x - P_W(x)$ is orthogonal to the subspace W ($\langle e, w \rangle = 0 \quad \forall w \in W$), which is the geometric basis of the Orthogonality Principle.

Vector Projection Geometry



Example: Gram-Schmidt Orthonormalization

Apply the Gram-Schmidt process to the following set of vectors in $\mathbb{R}^3$ to obtain an orthonormal basis:

$$v_1 = [1, 1, 0]^T, \quad v_2 = [1, 2, 0]^T, \quad v_3 = [0, 1, 2]^T$$

Solution:

1. For v_1 :

$$u_1 = v_1 = [1, 1, 0]^T \Rightarrow \|u_1\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$q_1 = \frac{u_1}{\|u_1\|} = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right]^T$$

2. For v_2 :

$$\langle v_2, q_1 \rangle = (1 \times \frac{1}{\sqrt{2}}) + (2 \times \frac{1}{\sqrt{2}}) + 0 = \frac{3}{\sqrt{2}}$$

$$u_2 = v_2 - \langle v_2, q_1 \rangle q_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \frac{3}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1.5 \\ 1.5 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \\ 0 \end{bmatrix}$$

$$\|u_2\| = \sqrt{(-0.5)^2 + 0.5^2 + 0} = \sqrt{0.5} = \frac{1}{\sqrt{2}}$$

$$q_2 = \frac{u_2}{\|u_2\|} = \sqrt{2} \begin{bmatrix} -0.5 \\ 0.5 \\ 0 \end{bmatrix} = \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right]^T$$

3. For v_3 :

$$\langle v_3, q_1 \rangle = \frac{1}{\sqrt{2}}, \quad \langle v_3, q_2 \rangle = \frac{1}{\sqrt{2}}$$

$$u_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\|u_3\| = 2 \implies q_3 = \frac{u_3}{\|u_3\|} = [0, 0, 1]^T$$

The orthonormal basis is:

$$\left\{ \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Quick Revision — Module III

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Key Formulas at a Glance

Sign-Error LMS	$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \operatorname{sgn}(e(n))\mathbf{x}(n)$
Sign-Sign LMS	$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu \operatorname{sgn}(e(n)) \operatorname{sgn}(\mathbf{x}(n))$
Normalized LMS Update	$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\tilde{\mu}}{\ \mathbf{x}(n)\ ^2 + \epsilon} e^*(n) \mathbf{x}(n)$
Rank-Nullity Theorem	$\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim(V)$
Orthogonal Projection	$P_W(x) = \sum_{i=1}^k \langle x, q_i \rangle q_i \quad (q_i: \text{ orthonormal basis})$
Gram-Schmidt Step	$u_k = v_k - \sum_{j=1}^{k-1} \langle v_k, q_j \rangle q_j, \quad q_k = u_k / \ u_k\ $

One-Line Definitions

- **Normalized LMS:** A variant of LMS that divides step-size by input power to maintain stability.
- **Sign-Sign LMS:** A hardware-friendly LMS variant using only signs of data and error.
- **Inner Product Space:** A vector space equipped with a function defining angles and lengths (norms).
- **Gram-Schmidt Process:** An algorithm generating orthonormal bases from a set of linearly independent vectors.
- **Rank of Operator:** The dimension of the range space of a linear transformation.
- **Nullity of Operator:** The dimension of the null space of a linear transformation.

Frequently Asked / Viva Questions

Q1. What is the primary advantage of Sign-LMS algorithms?

They reduce hardware multiplier requirements. Sign-Sign LMS replaces the update multiplications with simple addition/subtraction, accelerating hardware processing.

Q2. Why does standard LMS sometimes diverge with speech inputs and how does NLMS solve it?

Speech has highly variable power. During high-amplitude segments, the fixed step-size of standard LMS can violate bounds ($2/\operatorname{Tr}(\mathbf{R})$), causing divergence. NLMS scales μ inversely with instantaneous power, ensuring stability.

Q3. State the Rank-Nullity Theorem.

For any linear operator $T : V \rightarrow W$ where V is finite-dimensional, $\dim(\mathcal{R}(T)) + \dim(\mathcal{N}(T)) = \dim(V)$, meaning Rank + Nullity = Dimension of domain.

Frequently Asked / Viva Questions (contd.)

Q4. Explain overlap-save block processing in FDAF.

It is a frequency-domain technique that computes block convolutions of input signals with filter weights using DFTs/FFTs, avoiding time-domain convolution overhead.

Q5. What is sub-band adaptive filtering and why is it used?

It divides a wideband signal into multiple narrower frequency bands. It speeds up convergence by reducing the eigenvalue spread (whitening) and lowers computational complexity by decimation.

Q6. Define the orthogonal complement of a subspace.

The orthogonal complement $W^\perp$ of a subspace W is the set of all vectors in the vector space that are orthogonal to every vector in W .

Q7. Write the formula for the orthogonal projection of vector x onto vector y .

$$P_y(x) = \frac{\langle x, y \rangle}{\|y\|^2} y.$$

Q8. What represents the inner product of two signals $x(n)$ and $y(n)$ in L_2 space?

$\langle x, y \rangle = \sum_{n=-\infty}^{\infty} x(n)y^*(n)$ (for discrete signals) or $\int x(t)y^*(t)dt$ (for continuous signals).

Q9. What is the regularization parameter ϵ in NLMS?

It is a small positive constant added to the denominator to prevent division by zero (or numerical instability) when the input vector norm $\|\mathbf{x}(n)\|$ is zero or very small.

Q10. How does the computational complexity of FDAF scale with filter length M ?

It scales as $O(M \log_2 M)$ compared to $O(M^2)$ for standard time-domain LMS.